Syeda Bisma

9356131746 | syedabisma173@gmail.com | GitHub | LinkedIn | X (Twitter) | Maharashtra, India

Profile Summary

Final-year Computer Science student with hands-on experience building supervised and unsupervised machine learning models, evaluated using metrics like Accuracy, Precision, Recall, F1-score, MSE/MAE, Gini, and Silhouette Score. Proficient in Python, SQL, Pandas, NumPy, Scikit-learn, with a strong foundation in Data Structures & Algorithms and DBMS. Experienced in building end-to-end deployed ML projects using Streamlit, from data preprocessing to model deployment. Seeking a Data Scientist/ML Intern role to apply statistical modeling and machine learning to solve real-world problems.

Technical Skills

Programming Languages: Python, SQL, C++, JavaScript
Frameworks & Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, NLTK, FastAPI
ML/DL Concepts: Regression, Classification, Clustering, Feature Engineering, Text Preprocessing, EDA
Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Spyder, Google Colab, Streamlit, Docker, Excel
Relevant Coursework: OOP, Data Structures & Algorithms, Statistics, Machine Learning, DBMS, Deep Learning

Projects

Loan Risk Prediction System | *Github*, *Website* 2026

- Built and deployed an end-to-end credit risk model to predict loan default probability, benchmarking four algorithms — Logistic Regression (76.0%) and XGBoost (96.2%) — selecting XGBoost as the production model for its stronger generalization on unseen data.
- Engineered credit-risk-specific features using WoE and IV to enhance predictive signal, and reduced multicollinearity via VIF analysis to improve model stability and interpretability.
- Tuned hyperparameters with GridSearchCV and Stratified K-Fold cross-validation to prevent overfitting on imbalanced classes, then applied SHAP (waterfall and beeswarm plots) to explain individual loan approval/rejection decisions to non-technical stakeholders.
- Deployed a production-ready Streamlit application with batch CSV prediction, downloadable results, and NeonDB (PostgreSQL) integration enabling real-time risk scoring for end users.
- *Technologies/Tools Used:* Python, SQL, PostgreSQL, Streamlit, XGBoost, SHAP

Bank Fraud Analysis | *Github* 2026

- Built a fraud detection pipeline on 550,000 banking transactions with 20 features to classify fraudulent activity (is_fraud), addressing severe class imbalance in the target variable.
- Compared Logistic Regression (99.09% accuracy) against Decision Tree (69.78% accuracy), selecting Decision Tree as the final model for capturing substantially more fraudulent transactions despite lower headline accuracy.
- Visualized decision rules via tree plots to support interpretability of fraud-flagging logic for non-technical stakeholders.
- *Technologies/Tools Used:* Python, Scikit-learn, Logistic Regression, Matplotlib

Banking Product Recommendation System | *Github*, *Website* 2026

- Built and deployed an unsupervised recommendation engine on banking customer data, using K-Means clustering and FP-Growth association rule mining to suggest relevant products based on behavioral patterns.
- Engineered a bias-aware feature pipeline by excluding demographic attributes from model training, relying solely on financial and behavioral signals.
- Reduced dimensionality of 30 behavioral features to 17 components via PCA, retaining 90% variance; validated cluster quality using Elbow Method and Silhouette Score.
- Applied percentile capping and log transformation to treat outliers and skew in financial features prior to modeling.
- Deployed a Streamlit application supporting real-time single-customer recommendations.
- *Technologies/Tools Used:* Python, K-Means, FP-Growth, joblib, Streamlit

Education

Dr. Babasaheb Ambedkar Technological University (DBATU) 2023 – 2027
B.Tech, Computer Science and Engineering *CGPA: 7.2*